

Adaptive Hybrid RAG Framework for Context-Aware Document Intelligence

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Abstract— Organizations and educational institutions accumulate large collections of reports, manuals, policies and research papers, but finding a specific piece of information in them is slow. Keyword search matches terms rather than meaning, and large language models (LLMs), although fluent, can produce inaccurate or unsupported answers when asked about private documents that were not part of their training data. This paper presents the design of an Adaptive Hybrid Retrieval-Augmented Generation (RAG) framework for context-aware document intelligence over PDF, DOCX and TXT files. The framework combines BM25 keyword retrieval with semantic retrieval based on Sentence-BERT embeddings indexed in FAISS, and an adaptive selector chooses BM25, semantic retrieval or both according to the query. Retrieved passages are ranked and assembled into a context from which an open-source LLM (Llama 3 or Mistral) generates the answer, and the passages used are highlighted in the original document so that the answer can be checked. Beyond single-document question answering, the framework performs cross-document reasoning, detects contradictory statements across sources, and estimates evidence strength from the sources that support an answer, with the aim of making document intelligence explainable and trustworthy. We describe the architecture, the planned implementation and an evaluation methodology covering retrieval, answer, evidence and cross-document measures. No experimental results are reported here; they will follow once the system has been evaluated.

Keywords— Retrieval-augmented generation, adaptive hybrid retrieval, BM25, semantic search, Sentence-BERT, FAISS, document question answering, contradiction detection, evidence strength, explainable AI.

I. INTRODUCTION

Universities and organizations keep much of their working knowledge in digital documents such as research papers, technical reports, manuals and policy files. As these collections grow, finding a specific answer usually means opening several files and reading through them by hand. Search over such collections is commonly keyword based: documents are ranked by how well their terms match the terms in the query. BM25 [1] is the standard example of this family, and it remains effective when the query uses the same words as the relevant passage. It cannot, however, connect a query about “leave rules” with a passage that speaks of an “absence policy”, because it has no notion of meaning beyond matching terms.

Semantic retrieval addresses this by mapping queries and passages to dense vectors, so that similarity is measured in an embedding space rather than by shared words [2], [3]. Sentence-BERT [3] produces sentence embeddings suited to this comparison, and libraries such as FAISS [4] make nearest-neighbor search over large numbers of vectors practical. Dense retrieval has its own weakness: it can return passages that are related to the topic of the query but do not contain the specific fact requested, and it may miss exact matches on names, identifiers or numbers [2]. Combining lexical and dense retrieval has been shown to improve retrieval effectiveness [5], [6], but the combination in those studies is fixed and does not depend on the query.

Retrieval-augmented generation (RAG) [7] connects retrieval to an LLM. Instead of relying only on what the model absorbed during training, the LLM receives retrieved passages together with the question and generates its answer from them. This matters for private documents, which the model has not seen and about which it may otherwise give fluent but inaccurate, or hallucinated, answers. Supplying retrieved text is intended to ground the answer, but it does not by itself allow the reader to verify it: the generated text may still contain statements that the retrieved passages do not support, and the user has no direct way to see which part

of which document an answer came from.

A further limitation appears when a collection holds several documents on the same subject, such as different versions of a policy or reports from different departments. These documents may agree, overlap or contradict each other. Question-answering tools that work on one document at a time can return an answer with a citation, but they do not indicate whether other documents disagree, or how many independent sources back the answer.

This paper proposes an Adaptive Hybrid RAG framework for context-aware document intelligence that addresses these problems together. The framework (i) combines BM25 and semantic retrieval and selects between them for each query, (ii) generates answers from the retrieved context with an open-source LLM, (iii) highlights the exact source text used, and (iv) analyzes the retrieved evidence across documents to detect contradictions and to estimate how strongly an answer is supported. We present the design of the framework, the planned implementation and an evaluation methodology; no experimental results are reported.

The rest of the paper is organized as follows. Sections II and III state the problem and the objectives. Sections IV and V review related work and identify the research gap. Sections VI and VII describe the proposed framework and its implementation, and Section VIII compares it with an existing system. Section IX describes the evaluation methodology, and Sections X–XIII cover expected contributions, limitations, future scope and conclusions.

II. PROBLEM STATEMENT

LLMs generate natural-language responses, but for private documents they may produce inaccurate or hallucinated answers, because the relevant content was not part of what they learned in training. What is needed is a document question-answering system that retrieves the relevant information from the documents a user uploads and returns an answer that is trustworthy and explainable, together with the evidence that supports it.

The setting is one in which organizations and educational institutions manage thousands of digital documents, including research papers, reports, manuals and policies. Finding specific information in such collections is time-consuming, and traditional keyword search does not capture the context or meaning of a user’s query.

Formally, let $D = \{d_1, \dots, d_n\}$ be a collection of documents in PDF, DOCX or TXT format, each divided into passages, and let q be a natural-language query. The task is to return an answer a to q ; a set of evidence passages E drawn from D on which a is based; the relations (agreement or contradiction) between passages in E that come from different documents; and an indicator of how strongly E supports a .

III. OBJECTIVES

The objectives of the project are as follows.

- 1) To develop an AI-powered document intelligence system for multiple document formats (PDF, DOCX and TXT).
- 2) To implement adaptive hybrid retrieval that combines BM25 and semantic search for accurate information retrieval.
- 3) To generate context-aware answers from the retrieved document content using an LLM (Llama 3 or Mistral).
- 4) To provide interactive evidence by highlighting, in the original document, the exact text used to generate the answer.
- 5) To support conversation history and document summarization for a better user experience.
- 6) To build a reliable and explainable document question-answering platform that minimizes hallucination and improves user trust.

Beyond these objectives, the framework extends to cross-document analysis, namely contradiction detection and evidence-strength estimation, which distinguishes it from conventional document question-answering tools.

IV. RELATED WORK / LITERATURE SURVEY

We reviewed ten works on retrieval-augmented generation, dense and lexical retrieval, hybrid retrieval and domain adaptation. For each we note its main contribution, its relevance to this project and the limitation that our framework is intended to address.

A. Retrieval-Augmented Generation

Lewis et al. [7] introduced RAG, which combines document retrieval with LLM-based answer generation so that outputs are conditioned on retrieved text. It is the basis of our generation stage. The work concentrates on retrieval and generation, with limited attention to explaining how an answer was derived and none to enterprise document collections. REALM [8] integrates retrieval into language-model pre-training, so that the model retrieves documents while it learns and gains better access to factual knowledge. Its computational cost is high and it is difficult to adapt to enterprise document collections, which is why we apply retrieval at inference time over a user’s own documents. Siriwardhana et al. [9] improve the performance of RAG models on domain-specific knowledge bases through domain adaptation. This is relevant because organizational documents are domain specific, but the study addresses adaptation rather than explainability, contradiction detection or evidence analysis.

B. Dense Retrieval and Similarity Search

Karpukhin et al. [2] introduced dense passage retrieval (DPR), which retrieves passages by embedding similarity for open-domain question answering. Our semantic retrieval

component follows the same principle, but DPR is a semantic-only method and may miss exact keyword matches. Sentence-BERT [3] uses Siamese BERT networks to produce sentence embeddings that can be compared for semantic similarity; it is the embedding model family we plan to use. It does not perform retrieval by itself and needs an external vector index. FAISS [4] provides such an index, with fast similarity search over millions of vectors, but it handles semantic search only and offers no keyword-based retrieval. Zhao et al. [10] survey dense retrieval, covering techniques, embedding models and evaluation methods. As a survey it does not propose a unified adaptive retrieval framework, but it is a useful reference for the choice of embedding models and evaluation measures.

C. Lexical and Hybrid Retrieval

Robertson and Zaragoza [1] describe the probabilistic relevance framework on which the BM25 ranking function is based. BM25 is our keyword retrieval component; because it relies on exact term matching, it cannot capture semantic relationships between different words or phrases. Wang et al. [5] show that BERT-based dense retrievers benefit from interpolation with BM25, which supports combining the two methods; their hybrid strategy is fixed and does not adapt to the query. Luan et al. [6] compare sparse, dense and hybrid representations for text retrieval, with a focus on retrieval accuracy rather than explainability or document intelligence.

V. RESEARCH GAP

The surveyed literature shows that the individual components of a document question-answering system are well studied, while several properties of the complete system are not. Table I lists the gaps we identified and the component of the proposed framework intended to address each.

The first two gaps concern retrieval: BM25 and dense retrieval are usually applied independently, and even where they are combined [5], [6] the combination does not depend on the query. The remaining gaps concern what happens after retrieval. A citation tells the user where a passage came from, but not whether the answer depends on one source or several, whether other documents disagree with it, or how the answer was derived. The framework described next addresses these gaps within one workflow rather than through separate tools.

VI. PROPOSED FRAMEWORK

The framework has an indexing stage, which prepares the documents once, and a query stage, which retrieves evidence, generates an answer and analyzes the evidence. Fig. 1 shows the architecture. Steps 1 to 6 form the indexing stage; Steps 7 to 9 cover retrieval at query time, and Steps 10 to 15 cover generation and evidence analysis.

A. Indexing Stage

Step 1 — Document ingestion. Users upload PDF, DOCX and TXT files, and several documents can be added to one collection. Each file receives a document identifier, and its metadata (file name, format and upload order) is kept so that answers can cite their sources and so that passages from different documents can be compared later.

Step 2 — Text extraction and preprocessing. Text is extracted with a parser for each format and cleaned by normalizing whitespace, repairing hyphenation at line breaks and removing repeated headers and footers. For PDF files the page number of each piece of text is retained. Because evidence highlighting depends on it, the position of every extracted passage in the original file (page and character offset) is stored.

Step 3 — Document chunking. The cleaned text is split into overlapping chunks that are short enough to embed and

TABLE I
RESEARCH GAPS IDENTIFIED IN THE LITERATURE AND THE CORRESPONDING COMPONENTS OF THE PROPOSED FRAMEWORK

No.	Gap identified in the literature	Corresponding component of the proposed framework
G1	BM25 and dense retrieval are mostly used independently of each other.	Adaptive hybrid retrieval that combines BM25 with FAISS-based semantic search.
G2	The retrieval strategy is not chosen according to the type of query.	Adaptive retrieval selector that uses BM25, semantic retrieval or both for each query.
G3	Limited explanation of how an answer was retrieved and derived.	Evidence highlighting and interactive navigation to the source text in the original document.
G4	Limited reasoning across multiple documents.	Cross-document reasoning over evidence from several uploaded documents.
G5	Contradictory information across documents is rarely identified.	Contradiction detection module that flags conflicting statements from different sources.
G6	Citations are provided, but the strength of support for an answer is not estimated.	Evidence-strength analysis indicating whether an answer rests on one source or on several.
G7	Emphasis on answering questions rather than on broader document intelligence.	A single workflow combining question answering, summarization, conversation history and evidence analysis.
G8	Focus on single-document or open-domain settings, with limited support for enterprise-style collections.	Support for multi-document collections in PDF, DOCX and TXT, with a modular design for organizational knowledge management.

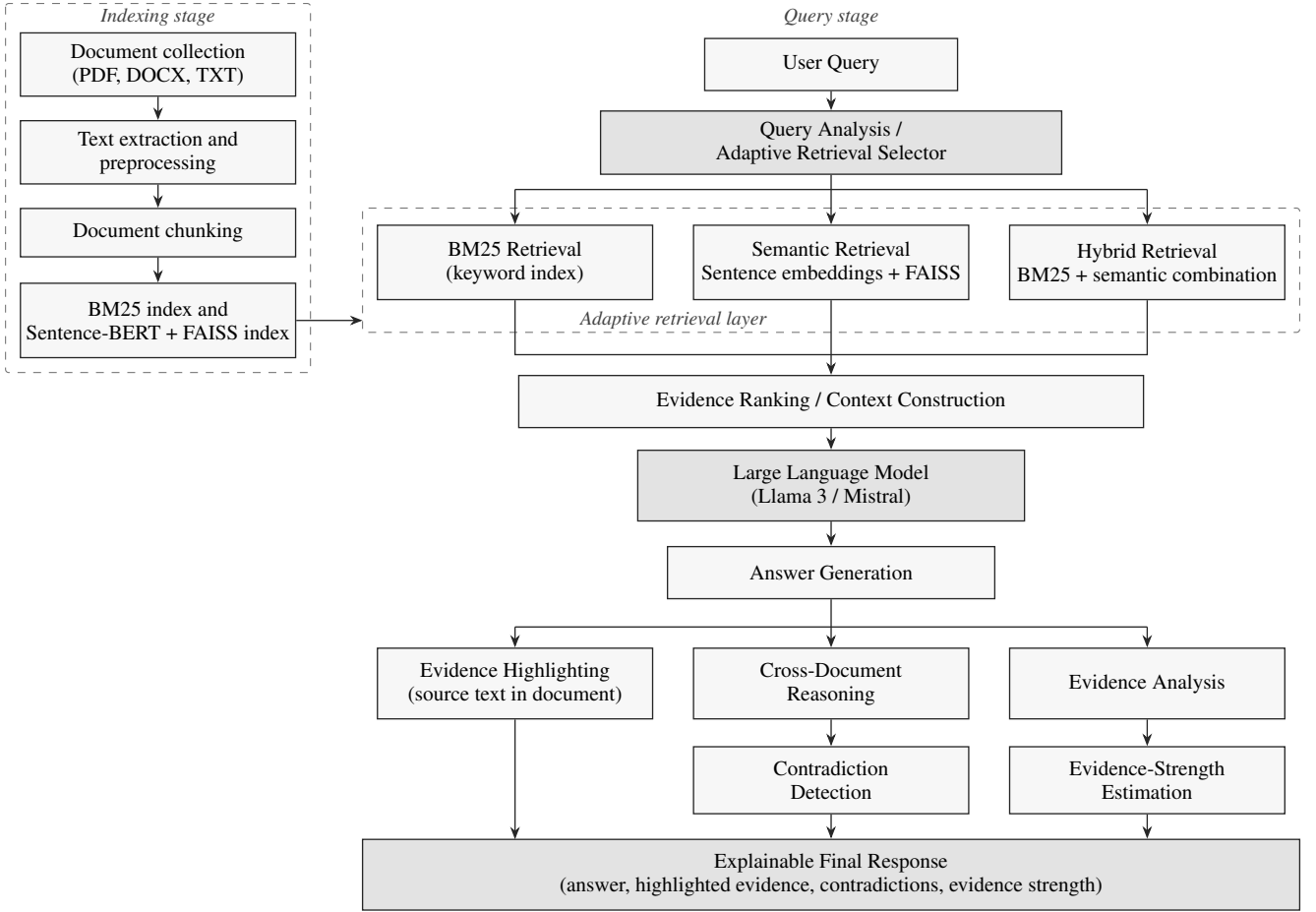


Fig. 1. Proposed architecture of the adaptive hybrid RAG framework.

to fit, several at a time, in the LLM context. Each chunk records its source document, page and character span. Chunk size and overlap are tunable parameters, to be set empirically, since retrieval quality depends on them (Section XI).

Step 4—BM25 indexing. The chunks are tokenized and a BM25 index [1] is built over them. For a query q and a chunk c , the score is

$$s_{\text{bm25}}(q, c) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, c) (k_1 + 1)}{f(t, c) + k_1 \left(1 - b + b \frac{|c|}{\text{avgl}}\right)} \quad (1)$$

where $f(t, c)$ is the frequency of term t in c , $|c|$ is the chunk length, avgl is the average chunk length, and k_1 and b are free parameters.

Step 5—Semantic embedding generation. Each chunk is encoded into a dense vector $\mathbf{e}_c$ with a Sentence-BERT model [3], and queries are encoded by the same model. Semantic similarity is the cosine of the two vectors,

$$s_{\text{sem}}(q, c) = \frac{\mathbf{e}_q \cdot \mathbf{e}_c}{\|\mathbf{e}_q\| \|\mathbf{e}_c\|}. \quad (2)$$

Step 6—FAISS vector indexing. The chunk embeddings are stored in a FAISS index [4], which supports nearest-neighbor search over the vectors. The BM25 index and the FAISS index use the same chunk identifiers, so that results from the two can be merged.

TABLE II
ILLUSTRATIVE QUERIES AND THE RETRIEVAL STRATEGY THE
SELECTOR IS INTENDED TO CHOOSE

Query property	Illustrative query	Strategy
Exact terms or identifiers	“notice period in Section 4.2”	BM25
Paraphrase of the document’s wording	“How are staff paid for time away from work?”	Semantic
Both exact terms and conceptual wording	“What does Section 4.2 say about time away from work?”	Both (hybrid)

The queries are examples for illustration and are not taken from an evaluation.

B. Query-Time Retrieval

Step 7—Adaptive retrieval selection. A query analysis step inspects the query and selects BM25, semantic retrieval or both. Queries that contain exact terms, such as names, identifiers, section numbers, acronyms or quoted phrases, are better served by lexical matching. Conceptual queries phrased differently from the document favor semantic retrieval, and queries with both properties use both. The selection can be formalized as an interpolation of normalized scores,

$$s(q, c) = \lambda_q \hat{s}_{\text{bm25}}(q, c) + (1 - \lambda_q) \hat{s}_{\text{sem}}(q, c), \quad (3)$$

where $\hat{s}$ denotes a score normalized to $[0, 1]$ over the candidate set and $\lambda_q \in [0, 1]$ is set by the selector: $\lambda_q = 1$ gives BM25 only, $\lambda_q = 0$ gives semantic retrieval only, and intermediate values give hybrid retrieval. How λ_q is determined (rules over query features, or a learned or LLM-assisted classifier) will be decided during implementation and compared against fixed- λ hybrid baselines, which correspond to the fixed strategy of Wang et al. [5]. Table II gives illustrative examples of the three cases.

Step 8—Evidence ranking. Candidates from the selected retriever or retrievers are merged by chunk identifier, ranked by the combined score, and the top k chunks are kept. Near-duplicate chunks are removed, and relevant chunks from different documents are retained so that later cross-document analysis has several sources to compare.

Step 9—Context construction. The top-ranked chunks are assembled into the LLM context with a label for each source document and, for follow-up questions, the conversation history. The prompt instructs the model to answer only from the supplied context and to say so when the context does not contain the answer.

C. Generation and Evidence Analysis

Step 10—LLM generation. An open-source LLM, Llama 3 [11] or Mistral [12], generates the answer from the constructed context through LangChain. The two models are interchangeable in the pipeline and can be compared in the evaluation.

Step 11—Evidence highlighting. The chunk positions recorded in Step 3 are used to locate the exact text in the original document. The passages that contributed to the answer are highlighted there, and the user can move from the answer to the highlighted passage to check it.

Step 12—Cross-document reasoning. When the retrieved evidence comes from more than one document, the passages are grouped by document and compared for what each says about the question. The aim is to identify where documents agree, where they complement one another and where they differ.

Step 13—Contradiction detection. Passages from different documents that address the same point are compared to decide whether they are consistent or in conflict. Conflicting statements are flagged and shown side by side with their

TABLE III
TECHNOLOGIES SELECTED FOR THE FRAMEWORK

Component	Role in the framework
Python	Main implementation language for document parsing, chunking, indexing, retrieval and the analysis modules.
LangChain	Connects the retrieved chunks to the LLM and handles prompt and context construction.
Sentence Transformers	Generates Sentence-BERT embeddings for document chunks and user queries.
FAISS	Stores the chunk embeddings and performs nearest-neighbor search for semantic retrieval.
BM25	Builds the keyword index over the same chunks and provides lexical ranking.
Llama 3	Open-source LLM that generates answers from the retrieved context.
Mistral	Alternative open-source LLM for the same task, allowing the two models to be compared.
FastAPI	Back-end service that connects the user interface to the retrieval and generation pipeline.
Streamlit	Web interface for uploading documents, asking questions and viewing answers with highlighted evidence.

sources. The comparison may be LLM assisted; the exact method will be fixed during implementation. A flag is meant to prompt review by the user, not to settle which statement is correct.

Step 14—Evidence-strength analysis. The framework estimates how strongly the retrieved evidence supports the answer, in particular whether it rests on a single source or is corroborated by several. Further signals may include the retrieval scores of the supporting passages and whether contradictions were found in Step 13. The result is presented as an indicator, for example limited, moderate or strong support, rather than as a probability; the exact scoring function remains to be defined.

Step 15—Final explainable response. The response presented to the user combines the answer, the source citations, the highlighted evidence, the agreements and contradictions found across documents, and the evidence-strength indicator. The exchange is added to the conversation history for follow-up questions, and document summarization is offered as a further request type in the same interface.

VII. SYSTEM IMPLEMENTATION

This section lists the technologies selected for the project and the intended role of each. The paper describes the design and does not report results from a completed system.

A. Technology Stack

Table III gives the role of each component in the framework.

B. Development Plan

Development is organized into eight phases: (1) literature survey and requirement analysis; (2) document processing and text extraction; (3) embedding generation and hybrid retrieval; (4) LLM integration; (5) the cross-document intelligence module; (6) explainable evidence navigation; (7) front-end and back-end development; and (8) testing, evaluation and documentation.

C. Feasibility

Technical. The framework is Python based and built from open-source libraries and models (Table III). **Operational.** It accepts common document formats (PDF, DOCX and TXT), and the interface is intended to need minimal technical knowledge; the system can be deployed with FastAPI and Streamlit. **Economic.** All components are open source, so no expensive

TABLE IV
FEATURE COMPARISON BETWEEN THE EXISTING SYSTEM AND THE PROPOSED FRAMEWORK

Feature	Existing system	Proposed framework
Question answering	Yes	Yes
Summarization	Yes	Yes
Source citations	Yes	Yes
Multiple documents	Limited	Yes
Cross-document QA	Limited/No	Yes
Contradiction detection	No	Yes
Evidence strength	No	Yes
Adaptive hybrid retrieval	No	Yes
Interactive evidence	Basic	Yes
Explainable intelligence	Limited	Yes

Existing-system entries are based on the case-study analysis of ChatPDF. Entries for the proposed framework describe planned functionality.

commercial software is required. *Implementation.* The architecture is modular, so each module can be developed and tested separately, which suits phased development within the project timeline and academic development in general.

VIII. EXISTING SYSTEM AND COMPARISON

A. Existing System: ChatPDF

As a case study we examined ChatPDF [13], an AI-powered document question-answering platform in which users upload PDF documents and interact with them in natural language. It uses RAG to retrieve relevant content from the uploaded documents and generates contextual answers with source citations. It also provides document summarization and conversational interaction, which makes it useful to students, researchers and professionals. We used it to understand how current AI-based document question-answering systems work in practice.

The system is designed mainly for interaction with a single document, and the limitations noted here concern functions beyond that goal. Support for cross-document reasoning is limited, contradictory information across documents is not analyzed, and evidence strength is not estimated from several supporting documents. The emphasis is on answering questions rather than on document intelligence across a collection. These are differences of scope, and they define the space that the proposed framework addresses.

B. Comparison

Table IV compares the features of the existing system with those planned for the proposed framework.

IX. EVALUATION METHODOLOGY

The project has not yet produced experimental results, so this section describes how the completed system is to be evaluated. The measures below are proposed evaluation measures; actual numerical results require implementation and experimentation and are not reported in this paper.

A. Experimental Setup

The evaluation will use a collection of PDF, DOCX and TXT documents (research papers, reports, manuals and policies) with a set of test questions whose relevant passages have been annotated. The set will include questions answered by a single document, questions that need several documents, and document pairs known to agree or to conflict. Four retrieval configurations will be compared: BM25 only, semantic retrieval only, fixed hybrid retrieval and adaptive hybrid retrieval. Llama 3 and Mistral will be compared as generators.

TABLE V
PROPOSED EVALUATION MEASURES

Measure	What is assessed
<i>Retrieval</i>	
Precision@K	Share of the top- K chunks that are relevant.
Recall@K	Share of the relevant chunks found in the top K .
MRR	Rank of the first relevant chunk, averaged over queries.
<i>Answer</i>	
Relevance	Whether the answer addresses the question.
Completeness	Whether all parts of the question are answered.
Groundedness	Whether the statements are supported by the retrieved context.
Unsupported claims	Statements not supported by any retrieved passage.
<i>Evidence</i>	
Evidence relevance	Whether the highlighted passages are relevant to the answer.
Evidence coverage	Whether the highlighted passages cover the claims in the answer.
Source consistency	Whether the highlighted text matches the cited source and location.
<i>Cross-document</i>	
Agreement detection	Whether documents that agree on a point are identified as agreeing.
Contradiction detection	Whether conflicting statements are flagged correctly (precision and recall on annotated pairs).
Multi-document reasoning	Correctness of answers that need information from more than one document.

B. Retrieval Evaluation

Retrieval will be evaluated with Precision@K, Recall@K and Mean Reciprocal Rank (MRR). For a query q with relevant set R_q and top- K retrieved list L_q^K ,

$$P@K = \frac{|L_q^K \cap R_q|}{K}, \quad R@K = \frac{|L_q^K \cap R_q|}{|R_q|}, \quad (4)$$

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}, \quad (5)$$

where Q is the set of test queries and rank_q is the rank of the first relevant passage for q .

C. Answer, Evidence and Cross-Document Evaluation

The remaining measures assess the generated answers, the evidence shown to the user and the cross-document analysis (Table V). Answer and evidence measures will be judged by annotators against the source documents with a written rubric. The agreement and contradiction measures compare the output of the system with annotated document pairs. Unsupported claims will be reported as the share of statements in an answer that no retrieved passage supports.

X. EXPECTED CONTRIBUTIONS

The framework is expected to contribute the following.

- **Adaptive hybrid retrieval.** Query-dependent selection between BM25, semantic retrieval and their combination, in place of a fixed retrieval strategy.
- **Cross-document intelligence.** Retrieval and analysis across several uploaded documents instead of one document at a time.
- **Contradiction detection.** Identification of conflicting statements across documents, shown alongside their sources.

- **Evidence-strength analysis.** An estimate of how strongly an answer is supported, based on the evidence from one or more sources.
- **Interactive evidence navigation.** Highlighting of the exact text in the original document that was used to generate the answer.
- **Explainable AI workflow.** A pipeline in which each answer comes with its evidence, its cross-document context and a strength indicator, so that the user can see how the answer was obtained.
- **Unified document intelligence.** Question answering, summarization, conversation history and evidence analysis in one modular framework rather than in separate tools.

Conventional document question-answering systems concentrate on producing an answer. The framework instead emphasizes transparency, reasoning and explainability, so that the user receives an answer and can also see how it was derived.

XI. LIMITATIONS

The framework has design-level limitations that will affect its behavior and that the evaluation should quantify. Retrieval depends on the quality of chunking: chunks that are too short lose context, chunks that are too long dilute both the embedding and the BM25 score, and a passage that spans a chunk boundary may be split. Text extraction from PDF files can be imperfect, particularly for tables, multi-column layouts and page headers, and scanned documents contain no extractable text and would need OCR, which is not part of the current design. The two retrieval methods have complementary weaknesses: dense retrieval can return passages that are semantically related to the query but imprecise, while BM25 can miss semantic relationships between different words. Adaptive selection reduces the effect of these weaknesses only if the selector chooses well.

Grounding in retrieved text does not remove the possibility that the LLM generates unsupported content. Evidence highlighting shows which passages were used, but it does not by itself guarantee that the answer is correct or that the passages are accurate. Contradiction detection is difficult when statements differ because of time, scope or definitions; two versions of a policy, for example, may both be correct for different periods. A detected contradiction should therefore be treated as a flag for review. Evidence strength indicates how much retrieved support an answer has and is not a measure of absolute truth, since several documents can repeat the same error. Finally, the framework has not yet been evaluated, and its behavior remains to be tested.

XII. FUTURE SCOPE

Several extensions follow from the limitations above. On the retrieval side, the adaptive selector can be improved, for example by learning the choice of strategy from query features, and a reranking stage can refine the candidate passages before generation. Domain-specific embeddings may improve semantic retrieval for specialized collections. On the document side, OCR would allow scanned documents to be processed, and better handling of tables and document layout would improve extraction from structured content. For attribution, claim-level citations would link each statement in an answer to its supporting passage, and source-version tracking would record which version of a document an answer is based on. The contradiction module can be strengthened with better contradiction analysis and temporal reasoning, so that statements from different dates or versions are not treated as

conflicts by default. Finally, the evaluation can be extended to compare local LLMs and to use larger benchmarks built from organizational documents.

XIII. CONCLUSION

This paper presented the design of an Adaptive Hybrid RAG framework for context-aware document intelligence. The framework combines BM25 keyword retrieval with semantic retrieval over Sentence-BERT embeddings and FAISS, and an adaptive selector chooses between the two, or uses both, according to the query. Retrieved evidence is passed to an open-source LLM in a retrieval-augmented generation pipeline, and the passages behind each answer are highlighted in the original documents, so that the answer is grounded in text the user can inspect. Cross-document reasoning, contradiction detection and evidence-strength analysis extend the pipeline from answering questions to analyzing what a collection of documents says on a subject, and together with evidence highlighting they are intended to make the results explainable. The work reported here is a design and an evaluation plan, and no performance results are claimed. The next step is to develop the modules over the planned phases and to evaluate them with the measures described in Section IX.

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